

CS173: Discrete Structures

Problem Set 1

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1. Prove each of the following statements without truth tables.

(a) Prove $p \rightarrow p$ is a tautology for any proposition p .

Proof. Let p be a proposition.

$$\begin{aligned} p \rightarrow p &\equiv \neg p \vee p && \text{(By Conditional Disintegration)} \\ &\equiv \top && \text{(By Complement)} \end{aligned}$$

$\therefore p \rightarrow p$ is a tautology. □

(b) Prove $p \rightarrow q \equiv \neg q \rightarrow \neg p$ for all propositions p, q .

Proof. Let p and also q be propositions.

$$\begin{aligned} p \rightarrow q &\equiv \neg p \vee q && \text{(By Conditional Disintegration)} \\ &\equiv q \vee \neg p && \text{(By Commutativity)} \\ &\equiv \neg(\neg q) \vee \neg p && \text{(By Double Negation)} \\ &\equiv \neg q \rightarrow \neg p && \text{(By Conditional Disintegration)} \end{aligned}$$

$\therefore p \rightarrow q \equiv \neg q \rightarrow \neg p$ for all propositions p, q . □

(c) Prove $\neg(p \rightarrow q) \equiv p \wedge \neg q$ for all propositions p, q .

Proof. Let p and also q be propositions.

$$\begin{aligned} \neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && \text{(By Conditional Disintegration)} \\ &\equiv \neg(\neg p) \wedge \neg q && \text{(By De Morgan's Law)} \\ &\equiv p \wedge \neg q && \text{(By Double negation)} \end{aligned}$$

$\therefore \neg(p \rightarrow q) \equiv p \wedge \neg q$ for all propositions p and also q . □

(d) Prove $(p \wedge (p \rightarrow q)) \rightarrow q$ is a tautology for all propositions p, q .

Proof. Let p and also q be propositions.

$$\begin{aligned}
(p \wedge (p \rightarrow q)) \rightarrow q &\equiv (p \wedge (\neg p \vee q)) \rightarrow q && \text{(By Conditional Disintegration)} \\
&\equiv ((p \wedge \neg p) \vee (p \wedge q)) \rightarrow q && \text{(By Distributivity)} \\
&\equiv ((\neg p \wedge p) \vee (p \wedge q)) \rightarrow q && \text{(By Commutativity)} \\
&\equiv (\perp \vee (p \wedge q)) \rightarrow q && \text{(By Complement)} \\
&\equiv (p \wedge q) \rightarrow q && \text{(By Identity)} \\
&\equiv \neg(p \wedge q) \vee q && \text{(By Conditional Disintegration)} \\
&\equiv (\neg p \vee \neg q) \vee q && \text{(By De Morgan's law)} \\
&\equiv \neg p \vee (\neg q \vee q) && \text{(By Associativity)} \\
&\equiv \neg p \vee \top && \text{(By Complement)} \\
&\equiv \top && \text{(By Domination)}
\end{aligned}$$

$\therefore (p \wedge (p \rightarrow q)) \rightarrow q$ is a tautology for all propositions p, q . □

(e) Prove $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$ for all propositions p, q, r .

Proof. Let p, q , and also r be propositions.

$$\begin{aligned}
(p \vee q) \rightarrow r &\equiv \neg(p \vee q) \vee r && \text{(By Conditional Disintegration)} \\
&\equiv (\neg p \wedge \neg q) \vee r && \text{(By De Morgan's law)} \\
&\equiv r \vee (\neg p \wedge \neg q) && \text{(By Commutativity)} \\
&\equiv (r \vee \neg p) \wedge (r \vee \neg q) && \text{(By Distributivity)} \\
&\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{(By Commutativity)} \\
&\equiv (p \rightarrow r) \wedge (q \rightarrow r) && \text{(By Conditional Disintegration)}
\end{aligned}$$

$\therefore (p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$ for all propositions p, q, r . □

2. Prove each of the following statements for all propositions φ, ψ, ξ .

(a) $(\varphi \rightarrow \psi), (\psi \rightarrow \xi) \vdash \varphi \rightarrow \xi$

Hypothetical Syllogism

Proof. Let φ, ψ, ξ be arbitrary propositions, and suppose $(\varphi \rightarrow \psi)$ and $(\psi \rightarrow \xi)$.
To prove $(\varphi \rightarrow \xi)$, by the deduction rule, it suffices to show $\varphi \vdash \xi$.

Assume φ .

By the assumption $(\varphi \rightarrow \psi)$, we have ψ by modus ponens.

By the assumption $(\psi \rightarrow \xi)$, we have ξ by modus ponens.

Thus $\varphi \vdash \xi$.

Therefore, by the deduction rule, $(\varphi \rightarrow \xi)$. □

(b) $\vdash \varphi \rightarrow \varphi$

Proof. Let φ be an arbitrary proposition.

To prove $(\varphi \rightarrow \varphi)$, by the deduction rule, it suffices to show $\varphi \vdash \varphi$.

Assume φ .

Then we have φ .

Therefore, by the deduction rule, $(\varphi \rightarrow \varphi)$. □

(c) $\vdash \top$

Proof. From part (b), we have $\varphi \rightarrow \varphi$ for any proposition φ .

From part 1(a) $\varphi \rightarrow \varphi \equiv \top$

Therefore, we have $\top$. □

(d) $\varphi, \psi \vdash \varphi \wedge \psi$

Conjunction Introduction

Proof. Let φ and ψ be arbitrary propositions. Assume φ and also assume ψ .

To prove $\varphi \wedge \psi$, we use reductio ad absurdum.

We will show $(\neg(\varphi \wedge \psi) \vdash \psi)$ and $(\neg(\varphi \wedge \psi) \vdash \neg\psi)$.

Assume $\neg(\varphi \wedge \psi)$.

$$\begin{aligned}\neg(\varphi \wedge \psi) &\equiv \neg\varphi \vee \neg\psi && \text{(By De Morgan's law)} \\ &\equiv \varphi \rightarrow \neg\psi && \text{(By Conditional Disintegration)}\end{aligned}$$

We now have $(\varphi \rightarrow \neg\psi)$ and φ from the initial assumption

Thus we have $\neg\psi$ by Modus Ponens. However, we originally assumed ψ , so there is a contradiction.

Therefore, by reductio ad absurdum, we conclude $\varphi \wedge \psi$. □

(e) $\varphi \wedge \psi \vdash \varphi$

Conjunction Elimination

Proof. Let φ and ψ be arbitrary propositions. Assume $\varphi \wedge \psi$.

To prove φ we use *reductio ad absurdum*. Assume $\neg\varphi$.

By conjunction introduction we have $(\varphi \wedge \psi) \wedge \neg\varphi$:

$$\begin{aligned} (\varphi \wedge \psi) \wedge \neg\varphi &\equiv \varphi \wedge (\psi \wedge \neg\varphi) && \text{(By Associativity)} \\ &\equiv \varphi \wedge (\neg\varphi \wedge \psi) && \text{(By Commutativity)} \\ &\equiv (\varphi \wedge \neg\varphi) \wedge \psi && \text{(By Associativity)} \\ &\equiv (\neg\varphi \wedge \varphi) \wedge \psi && \text{(By Commutativity)} \\ &\equiv \perp \wedge \psi && \text{(By Complement)} \\ &\equiv \perp && \text{(By Domination)} \end{aligned}$$

Thus we have $\perp$.

However, from part 2(c), we have $\top$.

Furthermore:

$$\begin{aligned} \top &\equiv \neg\varphi \vee \varphi && \text{(By Complement)} \\ &\equiv \neg(\neg(\neg\varphi) \wedge \neg\varphi) && \text{(By De Morgan's law)} \\ &\equiv \neg(\varphi \wedge \neg\varphi) && \text{(By Double negation)} \\ &\equiv \neg(\neg\varphi \wedge \varphi) && \text{(By Commutativity)} \\ &\equiv \neg\perp && \text{(By Complement)} \end{aligned}$$

We have both $\perp$ and $\neg\perp$, which is a contradiction.

Therefore, by *reductio ad absurdum*, φ holds. □